3 Project Initialization

3.1 Naming Conventions

A set of rules or guidelines for naming anything in the project:

- Database
- Schema
- Tables
- Store Procedures

Decide on **naming conventions, language** and **avoid using reserved words** as object words e.g., table. We going to use the snake case (lower snake): customer_info.

3.2 Table Naming Conventions

3.2.1 Bronze Rules

All names must start with the source system name, and table names must match their original names without renaming.

<sourcesystem>_<entity>:

- ***<sourcesystem>:*** Name of the source system (e.g., crm, erp)
- ***<entity>:*** Exact table name from the source system.
- Example: crm_customer_info = Customer information from the CRM system.

3.2.2 Silver Rules

All names must start with the source system name, and table names must match their original names without renaming.

<sourcesystem>_<entity>

- ***<sourcesystem>:*** Name of the source system (e.g., crm, erp)
- ***<entity>:*** Exact table name from the source system.
- Example: crm_customer_info = Customer information from the CRM system.

3.2.3 Gold Rules

All names must use meaningful, business aligned names for tables, starting with the category prefix.

<category>_<entity>:

- <category>: Describes the role of the table such as dim (dimension) or fact (fact table).
- <entity>: Descriptive name of the table, aligned with the business domain (e.g., **customers**, **products**, **sales**).

<u>Glossary of Category</u>

Pattern	Meaning	Example(s)
dim_	Dimension table	dim_customer, dim_product
fact_	Fact table	fact_sales
agg_	Aggregated table	agg_customers, agg_sales_monthly

3.3 Column Naming Conventions

3.3.1 Surrogate Keys

- A **surrogate key** is an artificial, system-generated unique identifier used in a database table instead of a natural key.
- It is used as the primary key to uniquely identify each record.
- It is usually an auto-increment number (e.g., 1, 2, 3…) or a UUID.
- A **primary key** is a role in a table that uniquely identifies each record. It can be made of one or more columns.

All primary keys in dimension tables must use the suffix **_key**.

<table_name>_key

- <table_name>: Refers to the name of the table or entity the key belongs to.
- _key: A suffix indicating that this column is a surrogate key.
- Example: customer_key = Surrogate key in the dim_customers table.

3.3.2 Technical Columns

- A technical column or attribute is a column in a database table that is not part of the business data, but is added for system, processing, or data-management purposes.
- It helps the database or analytics system operate more efficiently, track changes, or maintain data integrity.

All technical columns must start with the prefix **dwh_**, followed by a descriptive name indicating the column's purpose.

dwh_<column_name>

- Dwh: Prefix exclusively for system-generate metadata.
- <column_name>: Descriptive name indicating the column's purpose.
- Example: dwh_load_date = System generated column used to store the date when the record was loaded.

3.3.3 Stored Procedure

All stored procedures used for loading data must follow the naming pattern:

load_<layer>

- <layer>: Represents the layer being loaded, such as bronze, silver, or gold.